

Temporal Trends in Carbon Cycling

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and the members of the Carbon Temporal Synthesis Group (in no particular order):

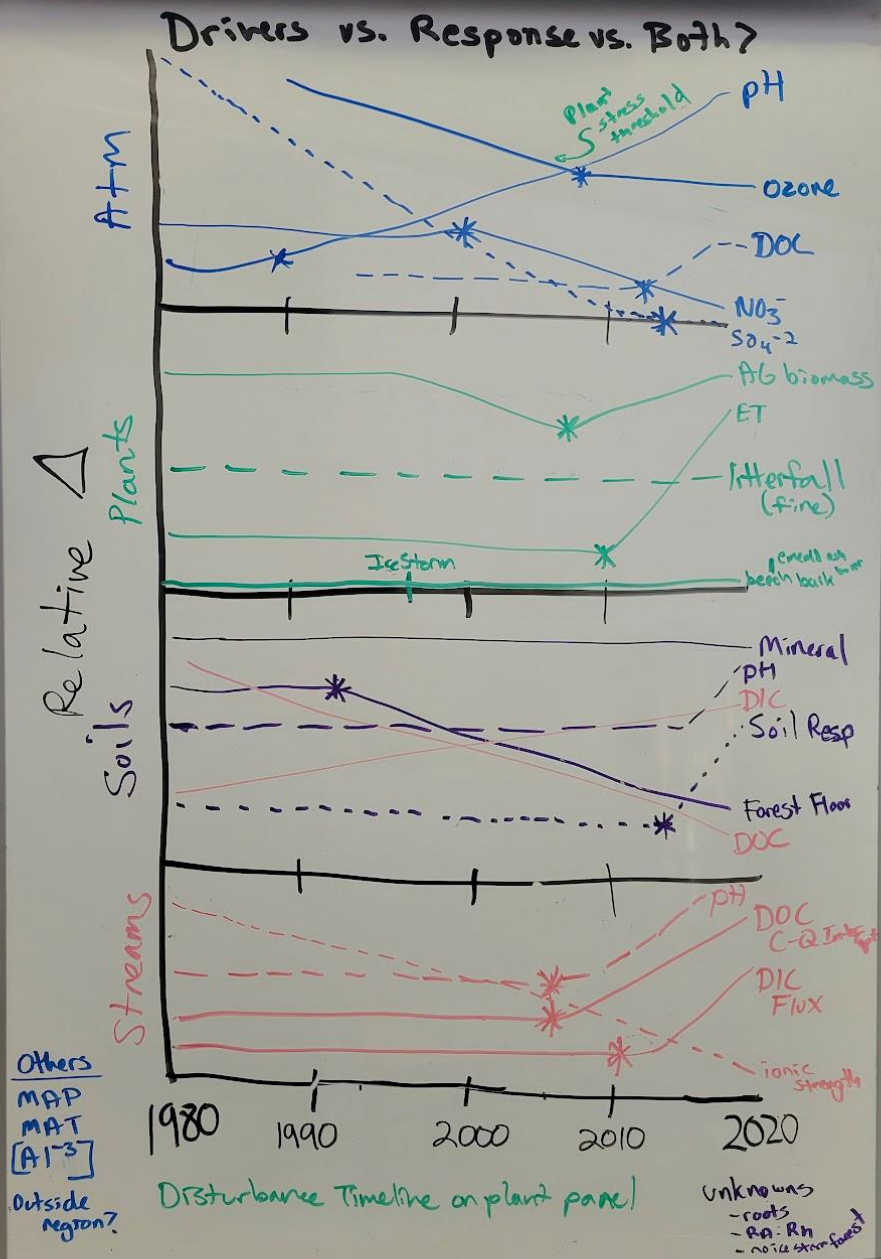
Sophie von Fromm, Angela Possinger, Chris Solomon, Andrew Reinman, Heili Lowman,
Peter Groffman, Emerson Conrad Rooney, Jason Demers, Geoff Wilson, Bob Fahey, Jeff
Garnas, Pam Templer, Darby Bergyl, Scott Olinger, Nina Lany, and Alix Contosta



How has global change affected the carbon cycle at Hubbard Brook?

Some Background

- This effort came out of the Carbon Cycle workshop in January
- We want to find a way to synthesize the myriad of long-term data products related to the carbon cycle
- There are many projects examining temporal trends at Hubbard Brook, but we **think** this is the only one focused on connecting all of the carbon cycle-related data



What is the rate and direction of change of carbon-related parameters in **streams**, **plants**, and **soils** over the past ~30-40 years at Hubbard Brook? Are these changes gradual or abrupt?

How do these trends compare to changes in drivers?

Preliminary Hypothesis:

There is an interaction wherein the responses of carbon fluxes (inputs and outputs) to climate change are more pronounced after a certain threshold of recovery from the effects of acid rain is reached.

Approach

Graph all of the potentially relevant datasets!



Standardized analyses of the temporal datasets:

1. Is it changing?
2. Is there a linear trend? Positive or negative? What is the (standardized) slope?
3. How it is nonlinear?
4. Does it have a breakpoint?

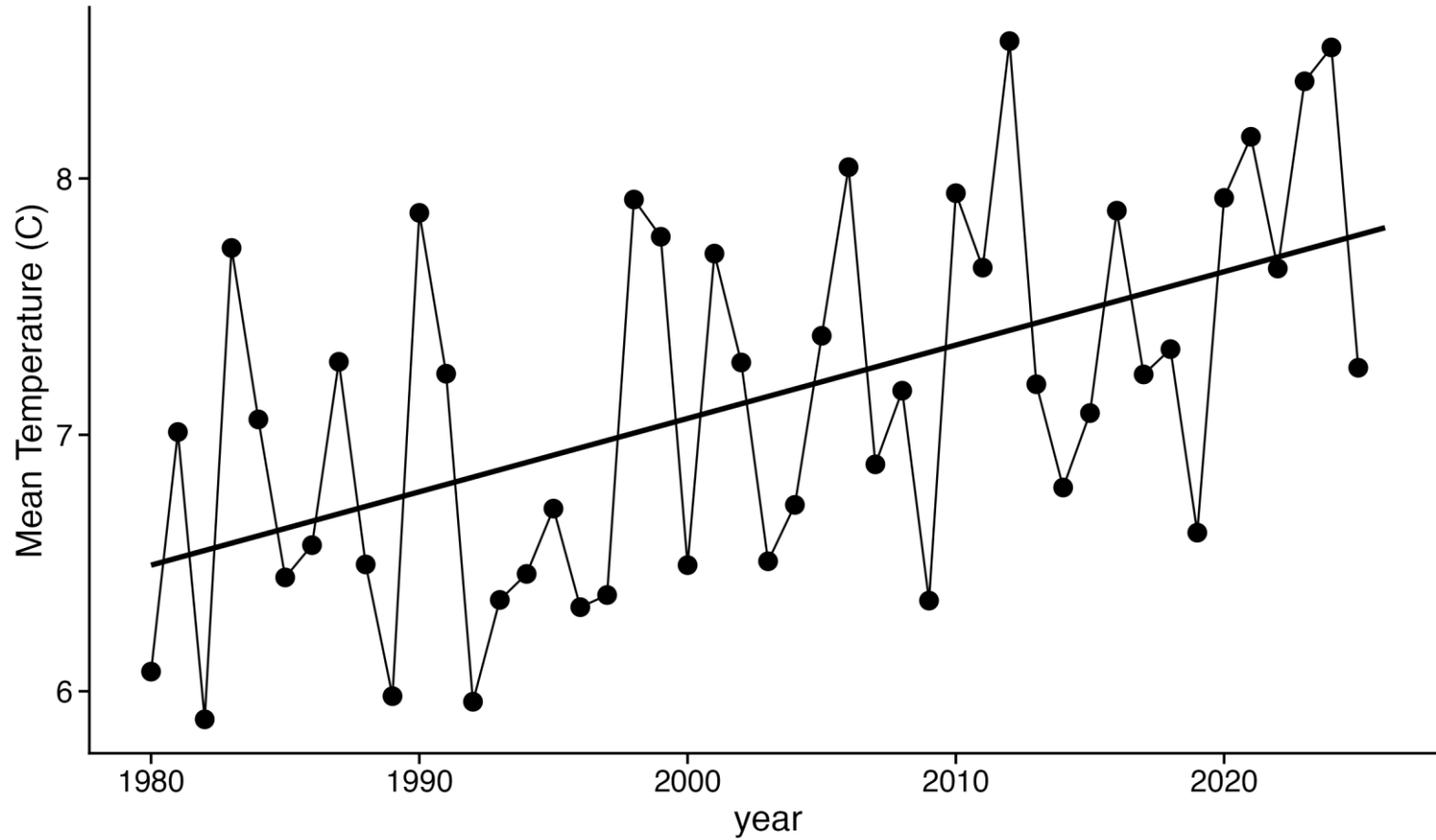
Unresolved:

1. Can we use comparisons between WS 1 and WS 6 to tease apart climate vs atmospheric chemistry drivers?
2. What about elevational gradients?

Climate Driver



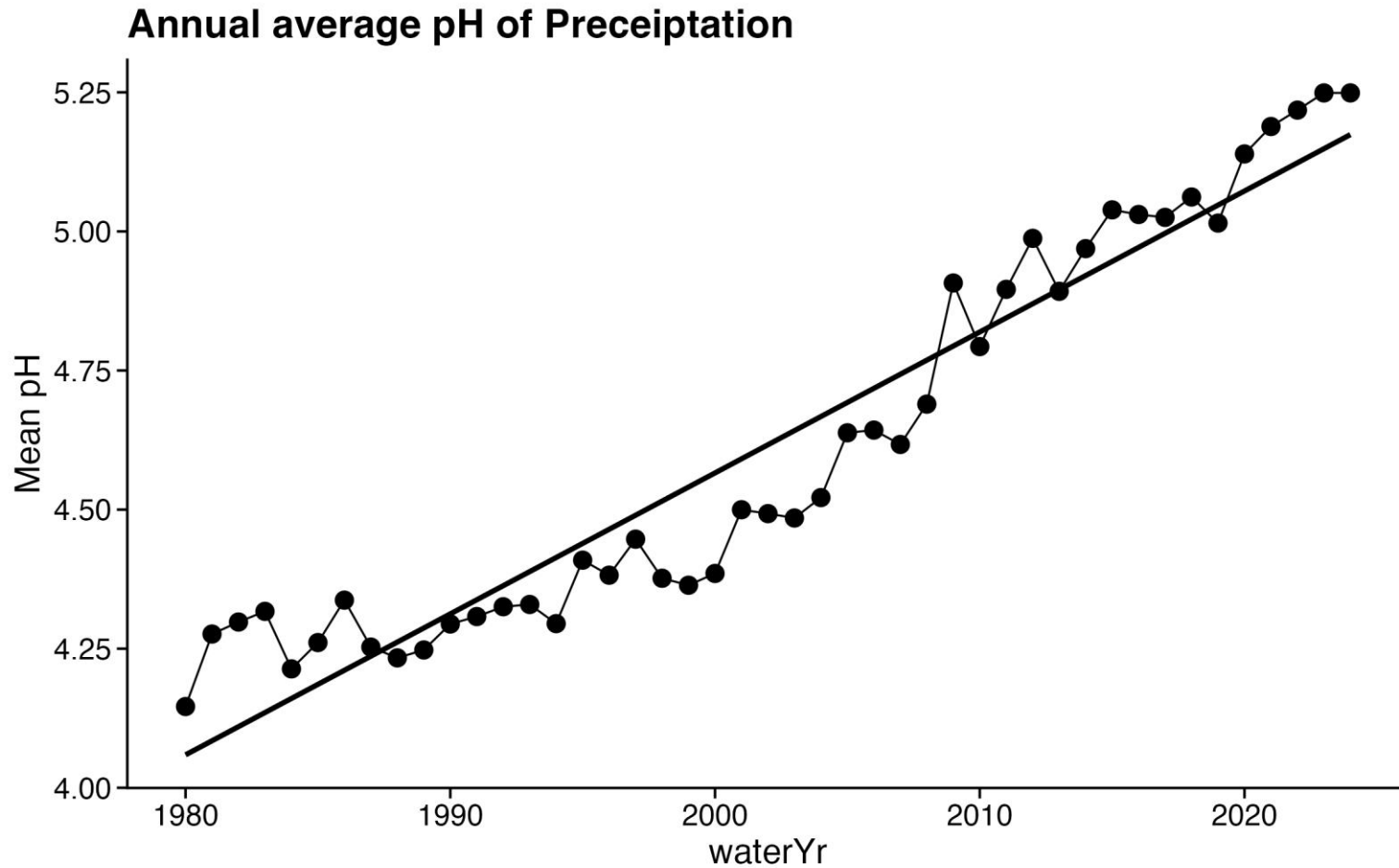
Annual average air temp at HQ



Slope = $0.0286 \text{ }^{\circ}\text{C year}^{-1}$

Standardized slope = 0.5352

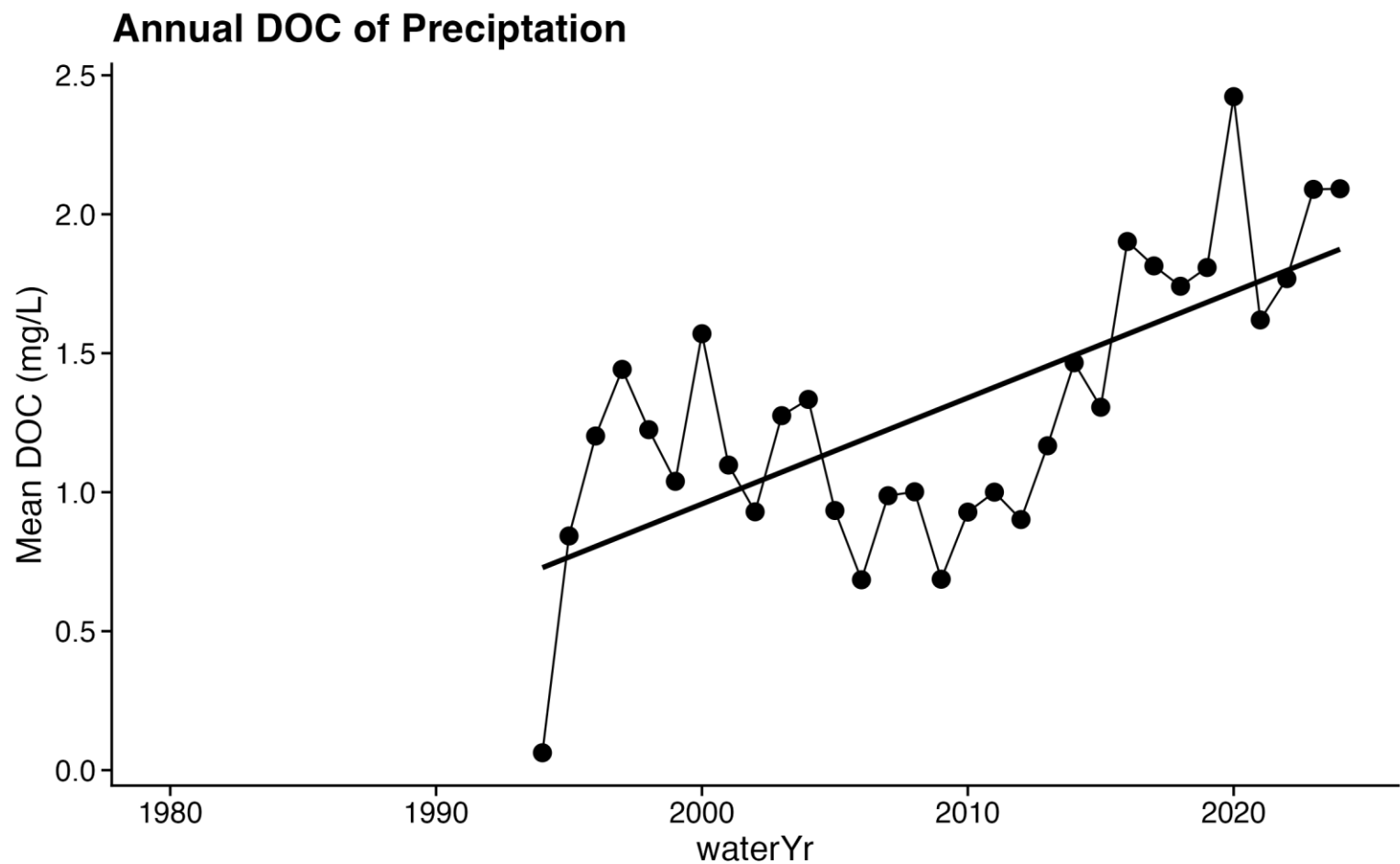
Atmospheric Chemistry Driver



Slope = 0.0253 pH units year⁻¹

Standardized slope = 0.9579

Atmospheric Carbon



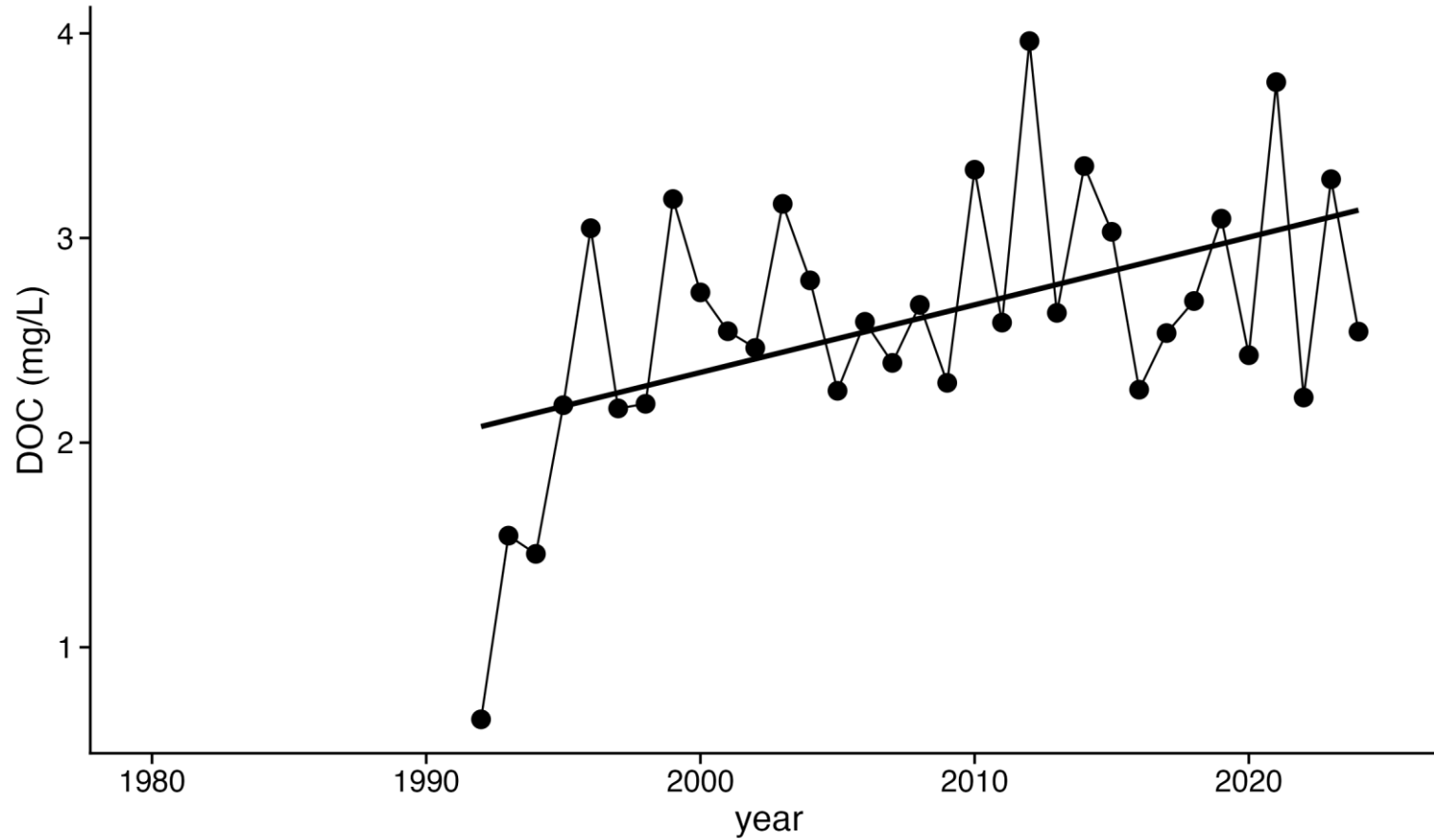
Slope = 0.03812 mg/L year⁻¹

Standardized slope = 1.001

Stream Response



Annual volume-weighted average stream DOC (W6)



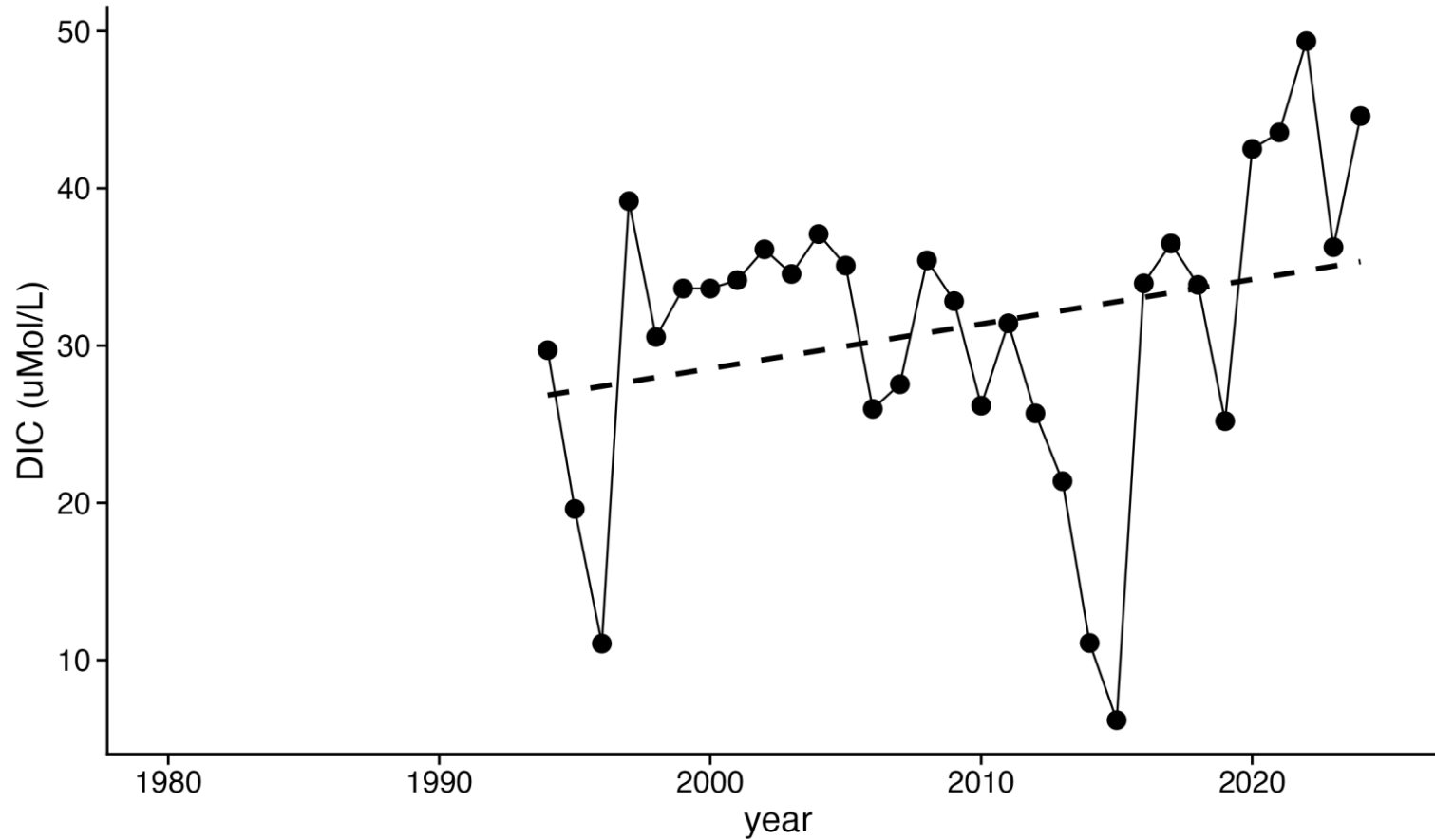
Slope = $0.0331 \text{ mg/L year}^{-1}$

Standardized slope = 0.6833

Stream Response



Annual volume-weighted average stream DIC (W6)



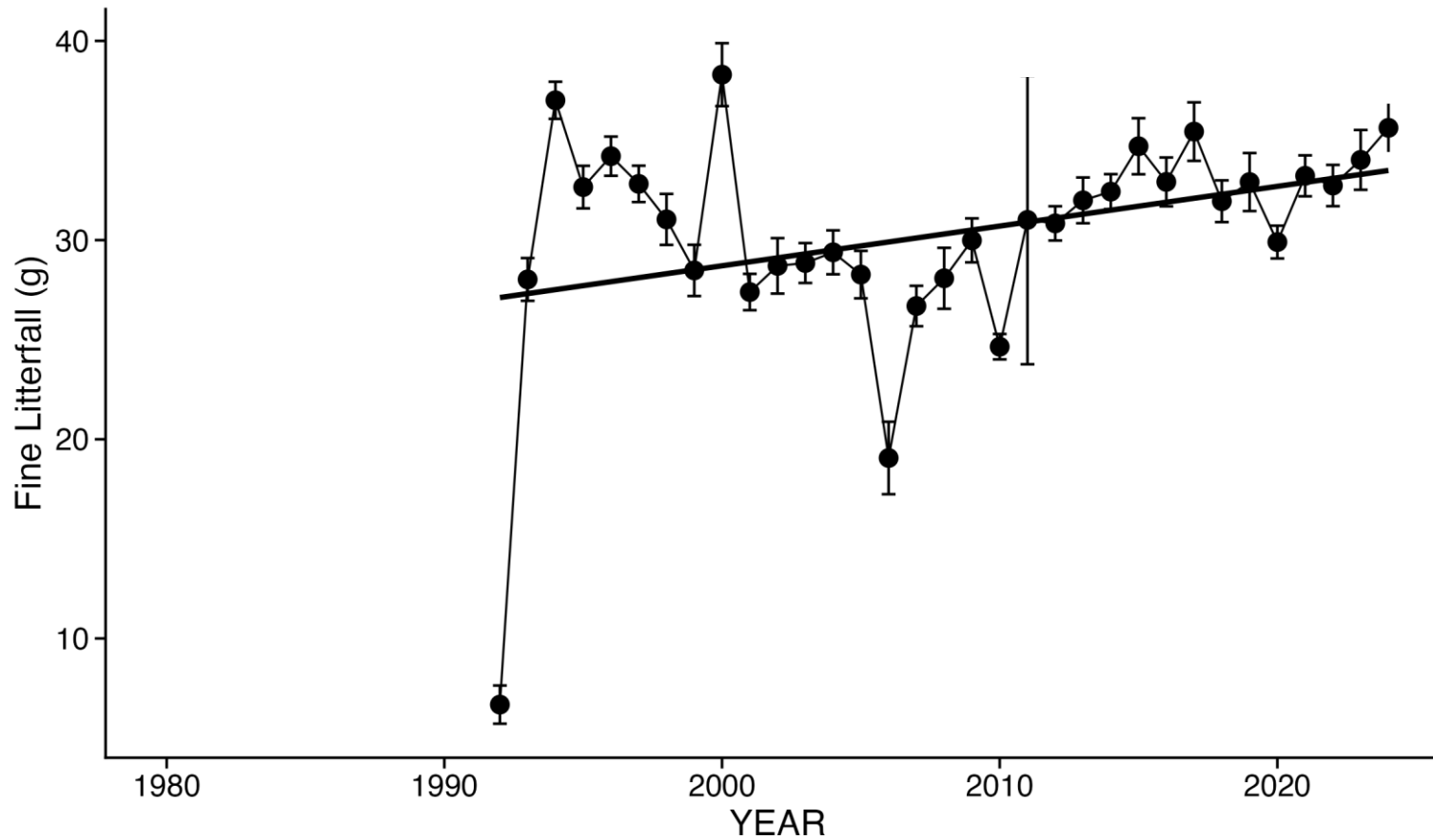
Slope = NS

Standardized slope = NS

Plant Response



Average Annual Litterfall (W6/BB)



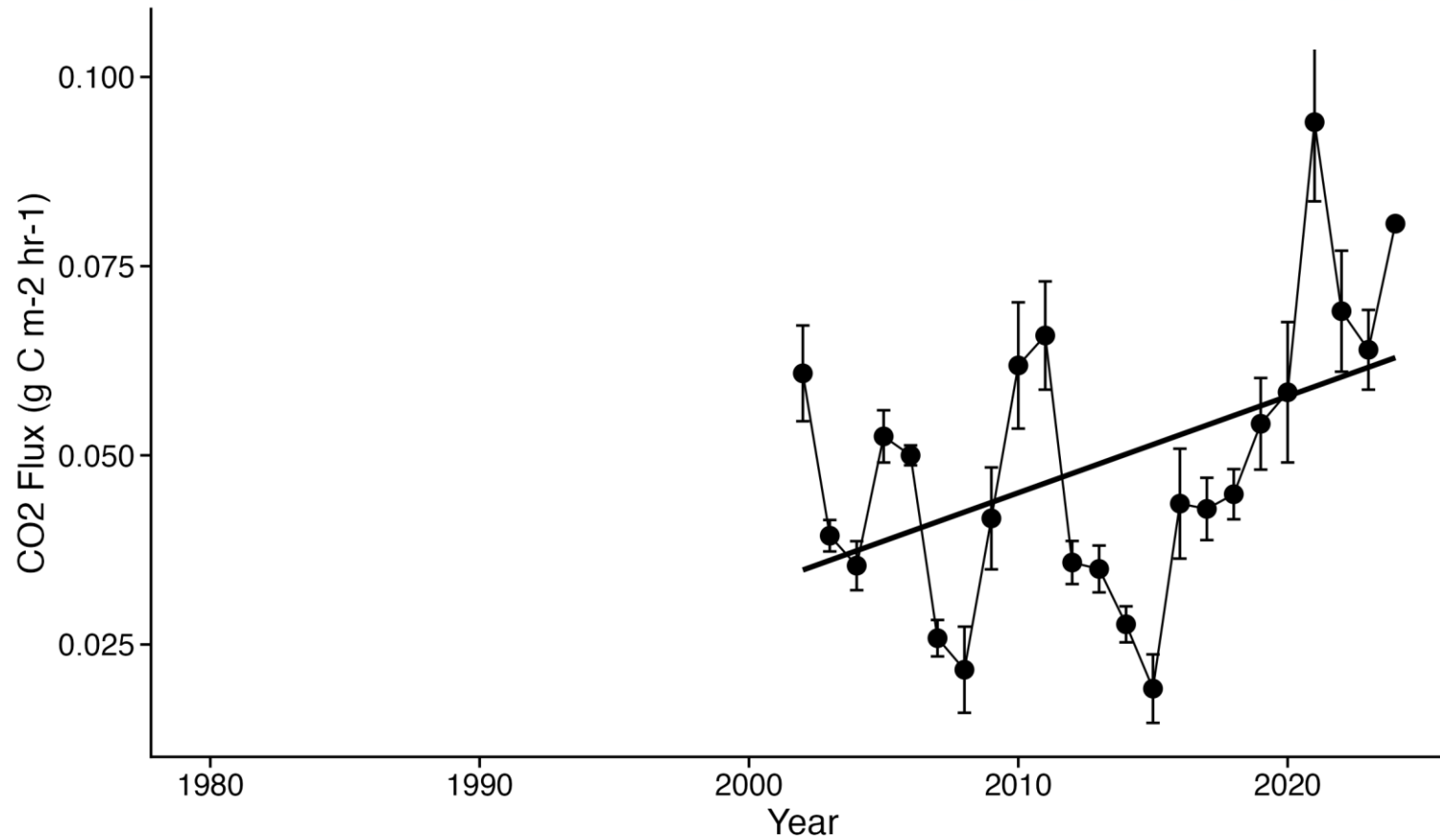
Slope = $0.1994 \text{ g year}^{-1}$

Standardized slope = 0.3406

Soil Response



Average Soil respiration (W6/BB)



Slope = 0.00128 g C m⁻² hr⁻¹ year⁻¹

Standardized slope = 0.4601

Preliminary Conclusions

- While not necessarily linear, almost all carbon flux-related parameters examined so far are increasing with similar standardized slopes
- Carbon data is relatively new by Hubbard Brook standards, so we only have about 20 years where all the data overlap
- There are strange values in some datasets, so expert review will be needed prior to standardized analyses
- Interested in participating? Thoughts? Concerns?
 - Please contact me: caitlin.hicks.pries@dartmouth.edu